Affordable solar-wind forecasting on WindNet's original holdouts

A compact experimental study with controlled ablations | 29 October 2026

Abstract. We tested inexpensive solar-wind forecasts using the original WindNet evaluation folds, without using solar images as predictors. Four hypothesis cycles led from short-history regression to recurrence features and a small ensemble. The frozen selection rule lowered mean root-mean-square error (RMSE) from **80.28 to 77.87 km/s** across five folds, but correlation fell from **0.550 to 0.515**. On Fold 5, which stayed sealed during model design, WindNet performed better. Feature ablations identify the 27-day recurrence band as the main useful addition. These findings support a cheap, transparent baseline, **not an established overall improvement over WindNet**.

1. Same evaluation rows, different inputs

We recovered the authors' released 211-Angstrom folds and retained all **2,198 forecast rows**. Every target matched the authors' saved WindNet predictions exactly; their mean RMSE and correlation reproduce the published 80.28 km/s and 0.55 [1, 2]. The forecast uses four days of recent input with a three-day delay (H = 4, D = 3). Our models replace images with daily OMNI speed history and optional plasma and magnetic measurements [3].

2. Main comparison

Method	Mean RMSE (km/s)	Mean r	Fold 5 RMSE (km/s)
Released WindNet	80.28	0.550	76.35
Four-speed ridge	87.68	0.217	81.83
Speed-recurrence ridge	79.45	0.484	79.60
Multivariate ridge	78.38	0.509	80.59
Residual neural ensemble	78.35	0.510	80.47
Convex ensemble	77.87	0.514	78.06
Frozen selection rule	77.87	0.515	78.06

Five-fold means are exploratory: Folds 1-4 informed model design. Correlations are averaged in Fisher-z space. Fold 5 is the primary comparison; lower RMSE is better.

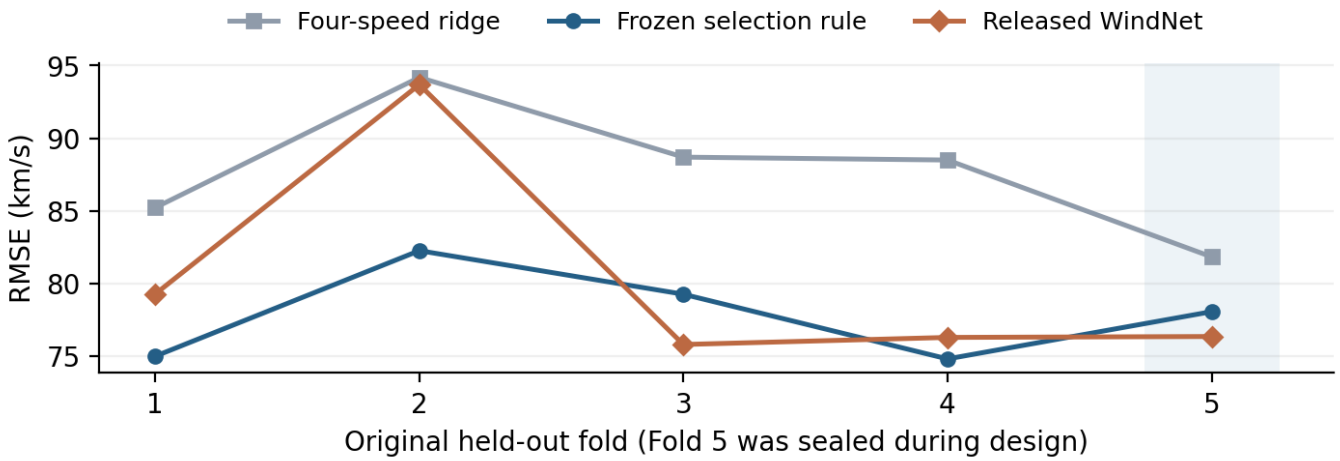


Figure 1. Error on each original held-out fold. The compact method has lower mean error, but the untouched Fold 5 favors WindNet (76.35 versus 78.06 km/s).

3. What actually helped?

The following additions keep the model family fixed to **ridge regression**. Each row uses the same held-out targets and training-only preprocessing, with its regularization penalty chosen inside the training folds. These requested ablations were planned after the final holdout was opened, so their interpretation is exploratory.

Feature additions	Mean RMSE (km/s)	Change from previous	Fold 5 RMSE (km/s)
Four recent speeds	87.68	-	81.83
Longer recent history only	88.14	+0.46	81.62
Add the 27-day recurrence band	80.24	-7.90	81.06
Add the 54-day band (all speed features)	79.45	-0.79	79.60
Add magnetic and plasma measurements	78.38	-1.07	80.59

A negative change is an error reduction. Extending recent history alone did not help. The 27-day band produced the largest incremental gain. Improvements in the five-fold average did not consistently carry over to Fold 5.

Remove one component from the full model

A complementary test removes one feature group from the full 78-column ridge model. Its reference RMSE is 78.38 km/s. This separates the 27-day and 54-day bands from the two measurement groups.

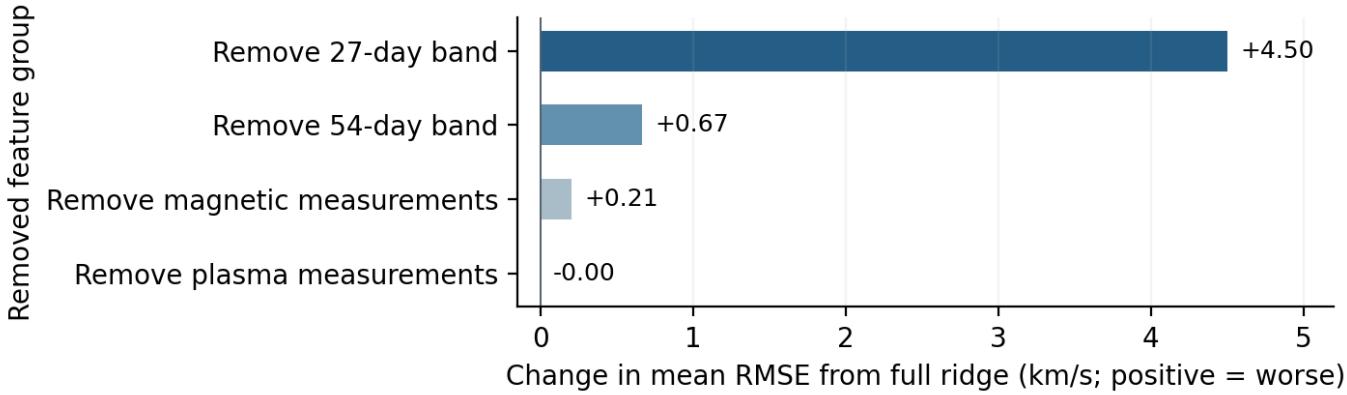


Figure 2. Mean error change when a feature group is removed. The 27-day band provides most of the measured benefit in this model family.

Controlled removal	Mean RMSE (km/s)	Change from full model	Fold 5 RMSE (km/s)
Remove 27-day band	82.88	+4.50	81.35
Remove 54-day band	79.05	+0.67	81.86
Remove magnetic measurements	78.58	+0.21	80.25
Remove plasma measurements	78.38	-0.00	79.87

Interpretation. Removing the 27-day band costs 4.50 km/s; removing the 54-day band costs 0.67 km/s. Magnetic measurements add only 0.21 km/s of mean benefit. Plasma measurements add essentially none: their removal changes RMSE by -0.0006 km/s. Neural and ensemble results in Table 1 are model-family comparisons, not clean feature-removal tests.

4. Hypothesis, experiment, observation

Step	Question tested	Observation
H1	Regularize four-speed regression.	88.78 km/s on development Fold 4; slightly better than fixed SVMs.
H2	Add recurrence and in-situ measurements.	74.83 km/s; a 15.72% reduction from H1.
H3	Add a GPU residual neural network.	74.47 km/s, only 0.90% below augmented ridge; the 2% criterion failed.
H4	Average different model errors.	73.85 km/s; passed the 1% gain criterion against the accepted tree rule.
E5	Freeze and evaluate original holdouts.	Lower five-fold mean error, but no confirmed advantage on untouched Fold 5.
E6	Remove feature groups one at a time.	27-day recurrence helps most; plasma adds essentially no average benefit.

H1-H4 used Folds 1-3 for fitting and Fold 4 for development; their scores are not independent test results. All six substantive experiments, including negative findings, were logged in RogNet/FixRelay. E6 is explicitly post-evaluation.

5. Evaluation safeguards and limits

The original release contains 157 blocks of 20 days, distributed across the five folds as 37, 25, 26, 20 and 49 blocks. We preserved the original 14 test windows per block (target offsets 6-19) and the training script's 13 windows. Repeated dates inside a fold were retained; no date crossed folds. All fitting lag values originating in withheld folds were masked before preprocessing. Missing values were handled inside training data only.

Speed inputs use lags 3-16, 24-31 and 51-58 days, alongside the original four recent speeds. Auxiliary measurements use lags 3-6. For each outer holdout, hyperparameters were selected on an inner training fold before refitting. The neural model used three fixed seeds. This changes the information available to the model relative to WindNet, so the comparison does not isolate an architectural advance.

On Fold 5, the compact method's RMSE was 1.71 km/s higher than WindNet's. A paired bootstrap with 5,000 resamples and 33 clusters of overlapping original blocks gave a 95% interval of **-6.16 to +7.95 km/s** for compact-minus-WindNet RMSE. The interval includes zero. The original WindNet code also selected checkpoints using its held-out fold. Random block splits and this small retrospective study cannot establish operational, chronological forecasting skill.

6. Reproduce and extend

The compact input file is about 265 KB. No raw SDO archive was downloaded and no image was used as a predictor; bundled processed images were discarded while recovering labels and fold identities. Main experiments ran on a Runpod 24 GB GPU slice, with estimated compute spending of about US$0.12 (billing had not settled). The temporary GPU and volume were deleted. Controlled ridge ablations ran locally. Code, exact predictions, split checks and experiment records are available in the public repository:

github.com/kahirhumibntwerag/windnet-low-cost-study

References

[1] Upendran, V., Cheung, M. C. M., Hanasoge, S., & Krishnamurthi, G. (2020). *Solar wind prediction using deep learning.* Space Weather, 18, e2020SW002478. doi:10.1029/2020SW002478.

[2] Authors' code and linked fold/prediction release: github.com/Vishal-Upendran/WindNet (revision a996baa). All 2,198 reference targets were matched exactly.

[3] NASA SPDF. *OMNIWeb data documentation.* omniweb.gsfc.nasa.gov/html/ow_data.html.